



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
Instruction Division

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
INSTRUCTION DIVISION

FIRST SEMESTER 2026-2027

COURSE HANDOUT(PART-II)

Date: 03-08-2026

In addition to Part-I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No. : BITS F463

Course Title : Cryptography

Instructor In Charge : Ankan Pal (ankan.pal@pilani.bits-pilani.ac.in)

Description: This course is an introduction to the basic theory and practice of cryptographic techniques used in information security. The course is intended to understand the mathematical concepts used for designing cryptosystems. The focus would be on the mathematical concepts from Number Theory and Abstract Algebra. We will delve into topics such as encryption (private-key and public-key), digital signatures, user authentication, key management, cryptographic hashing, Network security protocols, public-key infrastructure, digital rights management, and advancement in modern cryptology such as Identity Based Encryption, Steganography and Watermarking, Post-Quantum Cryptography, and Zero-Knowledge protocols.

Prerequisites: The course is self-contained, however a basic understanding of probability theory, complexity theory, and modular arithmetic from number theory would be helpful. The course is intended for advanced undergraduates and master students.

Course Objectives: Lectures deal with mathematical cryptography and related mathematical concepts required for modern cryptography and information security. The objective is to discuss and learn about basic tools and applications used in modern cryptography and to learn about some cryptographic protocols. On successful completion of the course students should be able to: understand basic principles and results of the theory of secure communication; know principles and problems of basic cryptosystems for encryption (both private and public key), digital signing and authentication; know methods to create core cryptographic protocols primitives; analyze and practically use simple cryptosystems; be understand basic principles of post-quantum cryptography and steganography.



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Course Learning Outcomes.

Upon successful completion of this course, students should be able to:

CLO1. Understand the fundamental principles and security requirements of modern cryptography.

CLO2. Utilize mathematical structures and ideas from Number Theory and Abstract Algebra to design and attack Cryptographic Protocols.

CLO3. Construct and Analyse various symmetric-key and public-key Cryptosystems.

CLO4. Reason and measure security of various Cryptosystems using computational and probabilistic techniques.

CLO5. Design and Analyse common cryptanalytic attacks and tools.

CLO6. Design secure communication mechanisms for the real-world using appropriate cryptographic primitives.

CLO7. Implement and Evaluate cryptographic algorithms with respect to security and computational efficiency.

CLO8. Understand modern developments in the area of Cryptography including zero-knowledge proofs and post-quantum cryptography.

Text Book.

[T1] B.A. Forouzan, D. Mukhopadhyay, Cryptography and Network Security, 3rd Edition, 2015, McGraw-Hill Education.



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Reference Books.

- [R1] W. Trappe, L.C. Washington, Introduction to Cryptography with Coding Theory, 2nd Edition, 2007, Pearson.
- [R2] S. Goldwasser, M. Bellare, Lecture Notes on Cryptography, 2008. Available online ([Link](#)).
- [R3] Douglas R. Stinson, "Cryptography Theory and Practice", Chapman & Hall/CRC, Third Edition, 2006.
- [R4] A.J. Menezes, P.C. van Oorschot, S.A. Vanstone, Handbook of Applied Cryptography, CRC Press, 1996.
- [R5] S. Goldwasser, M. Bellare, Lecture Notes on Cryptography, 2008.
- [R6] O. Goldreich, Foundations of Cryptography Volume 1: Basic Tools, Cambridge University Press, 2004.
- [R7] O. Goldreich, Foundations of Cryptography Volume 2: Basic Applications, Cambridge University Press, 2004.
- [R8] Bruce Schneier, Applied Cryptography: Protocols, Algorithms, and Source Code in C, Second Edition 1995, John Wiley & Sons.
- [R9] H. Delfs, H. Knebel, Introduction to Cryptography: Principles and Applications, 2nd Edition, 2007, Springer.
- [R10] S. Arora, B. Barak, Computational Complexity: A Modern Approach, 2009, Cambridge University Press.



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Course Plan.

Module	Broad Area	Lecture	Reference	Topics	Learning Objectives
Module I - Introduction, Motivation, and Background.	Art of Cryptography	L1-L5	T1-Ch 1	Objectives of Cryptography, Art of Cryptography, Security Overview (Goals, Attacks and Mechanisms), Classical Encryption Techniques - Historical Notes.	To understand the fundamental ideas, terminology, objectives and security requirements of cryptography; distinguish major cryptographic approaches and appreciate the role of mathematics, computation and randomness in secure communication.
				Basic Definitions (Encryption, Decryption, Keys, Cipher-Text, Plain-Text), Trade-off between Security and Speed (Standard Examples - Enigma Vs Lattice-based Cryptography), Motivation for Steganography and Watermarking.	
			T1-Ch 3, Sec. 10.1, 10.2	Public Vs Private Key Cryptography, Motivating Examples - Shift Cipher, Substitution Cipher, Affine Cipher, <u>Block Cipher</u> , <u>Stream Cipher</u> , RSA (Only Key Exchange), How financial institutions ensure safety of our credit/debit cards?	
			T1-Sec. 14.4, R1-Ch 13	<u>Zero Knowledge Proofs</u> (Introductory Concept). Poker and Secret Coin-Flipping Game.	
			T1-Appendix L	Role of Mathematics in Cryptography, Connections between Probabilistic Algorithms and the art of secret sharing. Motivation for Complexity Theory - How do we know that a Cryptographic Protocol is safe? How we efficiently communicate through noise channels without loss of information? - Motivation towards Error Correcting Codes.	



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Module	Broad Area	Lecture	Reference	Topics	Learning Objectives
Module II - <u>Mathematical Cryptography</u> .	Number Theory and Applications	L6-L11	T1-Ch 2, 9	Divisibility, Euclid's GCD Algorithm.	To develop the number-theoretic foundations of cryptography and apply modular arithmetic, prime-number theory and related algorithms to understand the construction and security of cryptosystems such as RSA.
				Congruences, Chinese Remainder Theorem, Modular Exponentiation Algorithm.	
				Fermat's Theorem, Euler's Theorem, RSA (Key Exchange) - Practical Example.	
				Quadratic Residues, Primitive Roots.	
				Prime Numbers, Primality Algorithms I.	
				Primality Algorithms II, The key issue to break RSA.	
	Algebraic Structures and Applications	L12-L16	T1-Ch 4	Groups, Subgroups, Is Group-based Cryptography possible? <u>Discrete Logarithm Problem</u> (Basic Concept).	To understand how groups, rings, fields and elliptic curves provide mathematical structures for cryptography, and apply these structures to key-exchange and public-key cryptosystems.
				Rings and Fields, Examples of Rings and Fields, Integer Ring, Polynomial Rings over Fields, Galois Fields, Field Extensions, Number Fields.	
			R1-Ch 17	Can Algebraic Structures be used for Cryptography (Number Fields, Lattices)?	
			T1-Sec. 10.5, R1-Ch 16	Elliptic Curves over Finite Fields. Group Structure.	
				Diffie-Hellman Key Exchange, Elliptic Curve Cryptosystems, Application to Bitcoin and Blockchain Technology.	
	<u>Complexity and Probabilistic Thinking</u>	L17-L22	R2-Sec 12.2	<u>Interactive Proofs</u> , Interactive Proof for Quadratic Non-Residuosity. Zero Knowledge Proof for Graph Isomorphism.	To understand computational and probabilistic notions of cryptographic security through interactive proofs, pseudorandomness, indistinguishability and standard attack model designing such as CPA and CCA.
			T1-Appendix E, K; R2-Ch 5,6	Function Family, Random Functions and Permutations. Pseudo-Random Functions and Permutations.	
				Pseudo-Random Permutations under CPA, Pseudo-Random Permutations under CCA, Birthday Attack.	
				Indistinguishability under CPA. Indistinguishability under CCA. Implementation of CCA and CPA.	





Pseudo-Random Bit Generators, Polynomial-Time Indistinguishability, Pseudo-Random Distribution, Pseudo-Random Generator, Blum-Blum-Shub Pseudo-Random Generator.

Module	Broad Area	Lecture	Reference	Topics	Learning Objectives
Module III - Applied Cryptography	Symmetric Key Cryptography	L23-L25	T1-Ch 3,5	Feistel Ciphers, Vigenère Cipher, Permutation Cipher, Cryptanalysis of Classical Ciphers.	To understand the design principles of symmetric-key cryptosystems, modes of operation and classical ciphers, and analyse their security against common cryptanalytic attacks.
			R1-Sec. 4.5,4.6,4.7	Modes of Operations in Symmetric Key Schemes ECB, CBC, CBCC, CTR, CTRC, OFB.	
				Attack on ECB and CBCC, Attack on any Deterministic, Stateless Scheme.	
	DES and AES	L26-L27	T1-Ch 6,7	Data Encryption Standard (DES) and Advanced Encryption Standard (AES).	To understand the architecture and operation of modern block ciphers through DES and AES, and study how cryptanalytic techniques such as differential cryptanalysis are used to evaluate their security.
			T1-Appendix N	Differential Cryptanalysis.	
	Public Key Cryptography, RSA, and Cryptanalysis.	L28-L30	R2-Ch 2	Public Key Encryption Schemes, RSA Public Key Cryptosystem (Revision), Polynomial-Time Reductions, Rabin Function. Trapdoor One-Way Permutations. RSA Trapdoor, Polynomial-Time Indistinguishability.	To understand the mathematical and computational foundations of public-key cryptography and analyse RSA and related constructions using one-way functions, trapdoor functions, and design potent cryptanalytic attacks on RSA.
				Extracting Partial Information from the RSA Function, Low Exponent Attack on RSA. Rabin's Public Key Cryptosystem.	
				Goldreich-Levin Construction of a Hard-Core Predicate. Bit Security of RSA. One-Way Predicates, A Set of Trapdoor Predicates based on the RSA Assumption. Encrypting Single Bits using Trapdoor Predicates,	



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				Encrypting Single Bits using Hard-Core Predicates, Efficient Probabilistic Encryption.	
	Hashing	L31-L32	T1-Ch 11,12, Appendix E	One-Way Functions, Examples of One-Way Functions, Secure Hash Algorithm (SHA), Strong One-Way Functions, Weak One-Way Functions.	To understand the principles of cryptographic hash functions, collision resistance, and analyse collision-finding and birthday-based attacks.
				Collision Resistant Hash Functions, Collision Finding Attacks, Random-Input Collision Finding Attack, Birthday Attack, Collision-Resistance under Hidden-Key Attack.	

Module	Broad Area	Lecture	Reference	Topics	Learning Objectives
Module IV - Information Architecture, Security, and Recent Developments in the Field of Cryptography	Message Integrity and Secure Communication	L33-L35	T1-Ch 11	Message Authentication Scheme, Message Authentication Code (MAC), MAC Security, PRF MACs, Examples of Forgeable MACs.	To understand how cryptographic primitives provide message integrity, authentication and secure communication, and study their use in practical protocols such as PGP, SSL and TLS.
			T1-Ch 16	Introduction to Application Layer Security (PGP).	
			T1-Ch 17,18	Introduction to Transport Layer Security (SSL and TLS), Introduction to Network Layer Security.	
		L36-L38	R2-Sec 12.4	Identity based encryption, Homomorphic Encryption, Secure Multi-Party Communication, Electronic Elections.	To understand the design and security of digital-signature and



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	<i>Digital Signatures and Multi-Party Communication</i>		T1-Ch 13	Digital Signatures, Digital Signatures using the Trapdoor Function Model, Diffie-Hellman Digital Signature Scheme, RSA Digital Signature Scheme. El Gamal's Digital Signature Scheme. Rabin's Digital Signature Scheme, Attacks against Digital Signatures.	authentication schemes and explore advanced mechanisms for secure interaction among multiple parties.
			T1-Ch 14,15	Key Distribution, Entity and Message Authentication, Password-based Authentication, Challenge-Response based Authentication, Types of Forgery, Security Definition of Digital Signatures.	
	<i>Advanced Topics</i>	L39-L40	R1-Ch 19	Introduction to Quantum Computation, Motivation for Post-Quantum Cryptography, Shor's Algorithm, Why RSA is not safe anymore?	To understand the impact of quantum computation on classical cryptography and develop an introductory understanding of post-quantum approaches for designing quantum-resistant cryptosystems.
				Post-Quantum Cryptography, Can we use Elliptic Curves and Lattices for designing quantum-resistant cryptosystems?	



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Evaluation Plan.

Component	Details	Duration	Weightage (%)	Date and Time
<i>Continuous Evaluation (30 %)</i> <i>Note: 2 Best Scores out of the 4 Quizzes will be chosen for final evaluation.</i>	Quiz-Q1 (Closed Book)	0.5 Hour	10	TBA
	Quiz-Q2 (Closed Book)	0.5 Hour	10	TBA
	Quiz-Q3 (Closed Book)	0.5 Hour	10	TBA
	Quiz-Q4 (Closed Book)	0.5 Hour	10	TBA
	Programming Assignment	NA	10	TBA
<i>Mid-Semester Examination (30 %)</i>	Mid-Semester Examination (Closed Book)	1.5 Hours	30	07-10-2026, FN1
<i>Comprehensive (40 %)</i>	Comprehensive Examination (Open Book)	3 Hours	40	08-12-2025, FN





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Chamber Consultation Hour: 17:00-18:00, Friday, 2223-F, 1st Floor, FD-2.

Make-Up Policy: No makeup will be given for the quizzes. Make-up for other components will be given **only** in genuine cases of absence as per institute rules. *Prior permission from I/C is a must for makeup of any evaluation component.*

NC Criteria: Minimum of 10 % of the total marks of the course.

Attendance Policy: Students are expected to attend all the lectures and tutorials. The attendance is not linked to any of the academic evaluation components.

Notices: All announcement related to the course will be done by sending broadcast or personal mails.

Note: A course seminar will be organized as part of the course to provide industry exposure. The seminar will be about *Man-In-the Middle Attack and SQL Injections* (common attacks on Apps) by an Industry Expert. The time and venue for the seminar will be announced during the lectures.

Ankan Pal

Instructor-In-Charge



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